

Alvaro Perdomo Gutierrez

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across healthcare, fintech, and e-commerce, building scalable full-stack systems in Python, Django, and React with cloud infrastructure and observability. Experienced in distributed platforms, ML integration, modernization initiatives, and regulatory-compliant delivery using PostgreSQL, AWS, Docker, and Kubernetes.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Backend: Django, Django REST Framework, FastAPI, Flask, Node.js, Express.js

Frontend: React, Next.js, Redux, React Query, Webpack, Vite, Tailwind CSS, Material-UI

Database: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, TcaplusDB

Testing: Jest, Pytest, Cypress, Selenium, unittest, pytest-django

Cloud & DevOps: AWS, EC2, RDS, S3, Lambda, CloudFront, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Google Cloud Platform, Databricks

Others: RESTful API, GraphQL, Git, GitHub, CI/CD, Agile, GitHub Copilot, HIPAA, Grafana, Prometheus, Datadog, API Gateway, Nginx

EXPERIENCE

Leanware

May 2024–Present

Senior Software Engineer

- Led development of a healthcare SaaS platform delivering AI-driven clinical decision support, building the full-stack system in **Python**, **Django**, and **React** that processed patient data for Fortune 500 healthcare organizations while maintaining strict **HIPAA** compliance and audit logging.
- Drove modernization of the legacy monolith by migrating authentication and API services from Flask to **Django REST Framework**, refactoring database queries to use **PostgreSQL** connection pooling, and eliminating N+1 query patterns that had been causing page load latency above 2 seconds.
- Championed adoption of **Docker** and **Kubernetes** across the product infrastructure, containerizing all backend services and implementing auto-scaling policies that reduced cloud spend by **18%** while maintaining sub-100ms P95 response times under peak traffic.
- Implemented a scalable job processing system using **Celery** and **Redis** for asynchronous clinical data ingestion and model evaluation, handling 50K+ daily tasks and reducing synchronous API timeout incidents from 8–12 per week to near zero.

- Redesigned the frontend component library by consolidating **React** and **Material-UI** patterns across 12+ patient-facing dashboards, improving component reusability by **40%** and reducing bundle size from 450KB to 320KB through tree-shaking and code splitting.
- Established end-to-end test coverage for critical clinical workflows by writing **Pytest** integration tests on the backend and **Cypress** E2E tests on the frontend, catching a subtle insurance eligibility bug before production that would have affected 1,200+ members.
- Collaborated with the ML engineering team to integrate a precision-matching model trained on **Databricks**, exposing predictions through **REST APIs** and monitoring inference latency and accuracy drift using **Prometheus** and **Grafana** dashboards.
- Mentored three junior engineers on full-stack development practices, including code review standards, **Git** workflows, and debugging production issues in **AWS** cloud environments using **CloudWatch** logs and **X-Ray** tracing.
- Optimized database performance by adding strategic indexes to frequently filtered **PostgreSQL** tables and implementing a **Redis** caching layer for member eligibility lookups, reducing database CPU usage by **25%** during peak hours.
- Automated the release pipeline by enhancing **GitHub Actions** workflows to run unit tests, integration tests, and security scanning in parallel, cutting deployment time from 40 minutes to 12 minutes and enabling daily releases.
- Drove cross-functional collaboration with customer engineering teams to gather requirements for a new care navigation module, translating business needs into technical specifications and delivering a production-ready feature within a single sprint.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Spearheaded integration of an AI-powered visual recommendation system into the e-commerce platform using **Python**, **Django**, and **React**, processing 275K+ garment images per week and driving a **15%** increase in sell-through rate and **2.8%** improvement in iOS conversion.
- Owned the migration of the ML training platform from on-premise infrastructure to **Databricks**, enabling data scientists to deploy garment measurement models in 1 month compared to 2 quarters in the legacy system, accelerating MVP iteration from 2 months to 2 weeks.
- Extended the **REST API** surface to expose visual search and style recommendations to the mobile app, implementing request rate limiting and caching strategies with **Redis** to handle 100K+ image processing jobs daily without exceeding AWS cost budgets.
- Refactored the resale pricing service from a monolithic **Flask** application to microservices using **Django REST Framework**, splitting out inventory, pricing algorithm, and ranking concerns across separate **Kubernetes** pods that could scale independently during peak demand.
- Improved observability across the ML inference pipeline by instrumenting **Python** services with structured logging and **Prometheus** metrics, catching a model drift issue within hours of deployment that prevented an estimated **\$120K** in mispriced inventory.

- Collaborated with product and customer success teams to launch an AI Shopping feature combining image search and conversational style chat, coordinating frontend development in **React** and backend inference serving to ship within a compressed 6-week timeline.
- Hardened security posture by implementing OAuth2 authentication across internal admin tools, adding request signing to **API Gateway**, and enforcing HTTPS-only data transmission, reducing unauthorized API access attempts by **94%**.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Built and scaled the precision-matching ML platform using **Python** and **Django** to support 6x growth reaching 2M+ members while preserving clinical validity through rigorous test coverage and validation workflows that prevented a single mismatched member case in production.
- Engineered the global Care Navigation platform to support dependent eligibility verification and jurisdiction-aware benefit matching across 15+ countries, requiring complex **PostgreSQL** schema design and multi-language localization that enabled Fortune 500 onboarding including PepsiCo, Instacart, and Bain.
- Scaled **HIPAA**-compliant infrastructure on **AWS** by architecting multi-tenant data isolation using **RDS** encryption, audit logging, and fine-grained **IAM** policies, which certified the platform for enterprise healthcare deployment and supported rapid customer acquisition.
- Developed operational ML systems for member dropout risk prediction and demand forecasting using **Python**, **Pandas**, and **scikit-learn**, feeding insights into the customer success team's proactive outreach and reducing churn by 12% quarter-over-quarter.
- Collaborated across engineering and clinical teams to design the dependent eligibility feature launched in September 2021, coordinating **REST API** contracts and **React** frontend components to ensure accurate benefit assignment across family members and dependents.
- Implemented comprehensive integration tests using **Pytest** to validate clinical matching logic across complex rule engines, ensuring member-to-clinician matches met clinical guidelines and regulatory requirements before production deployment.

Tencent Games

Nov 2014–Nov 2020

Software Engineer

- Contributed to the backend services powering Honor of Kings, a top-tier mobile MOBA that grew from launch to 100M daily active users by November 2020, building game server logic in **C++** and **Python** that handled real-time multiplayer synchronization across millions of concurrent players.
- Implemented anti-cheat systems and game integrity checks using kernel-level ACE anti-cheat technology, preventing unauthorized account access and protecting the competitive integrity of ranked gameplay for millions of players monthly.
- Developed game-hosting and player-matching infrastructure on **Google Cloud Platform** and **TcaplusDB**, a distributed NoSQL database handling 40M+ daily requests, which enabled sub-100ms matchmaking latency critical for real-time competitive gameplay.

- Built the Tencent Gaming Buddy emulator platform in 2018, providing a **Docker**-containerized desktop client that allowed PC players to engage with mobile titles, expanding the addressable player base by integrating with existing game ecosystems and cross-platform progression.
- Supported the globalization of Arena of Valor by adapting game servers and balance logic for international markets, coordinating with localization teams and regional publishers to ensure feature parity across **Kubernetes**-orchestrated deployments in multiple geographic regions.
- Collaborated with external partners including Krafton on PUBG Mobile infrastructure, integrating third-party services for social features and in-game purchases that scaled to support 100-player battle royale sessions on Unreal Engine 4 without server-side crashes.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014